

Theorem: For every finite undirected graph, the sum of all vertex degrees is equal to twice the number of edges.

Proof: (2 points) Universal declaration: Let G be an arbitrary finite undirected graph with m edges.

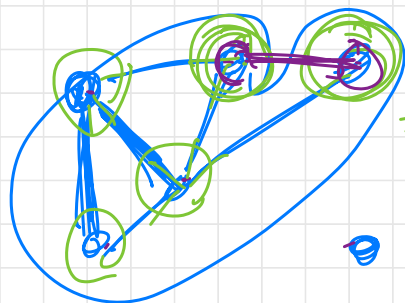
We induct on m , the number of edges in G .

(3 points) Induction hypothesis:

Assume that the claim holds for all smaller instances.

Assume that for all finite undirected graphs with $m' < m$ edges, the sum of all vertex degrees is $2m'$.

G'



$$2m = 2 \cdot 7 = 14$$

$$\begin{array}{r} 2+3+4+3+2 \\ = 14 \end{array}$$

$$m = 7 \text{ edges}$$

There are two cases:

- Base case: Suppose $m = 0$. Then every degree is zero, so the sum of the degrees is $0 = 2 \cdot 0$, so the claim holds.

- (5 points) Inductive case: Suppose $m \geq 0$.

Let G' be a graph formed by removing one edge from G , and let m' be # edge of G' .

$$\sum_{v \in G} \deg(v) = \sum_{v \in G'} \deg(v) + 2$$

$$= 2m' + 2$$

$$= 2(m-1) + 2$$

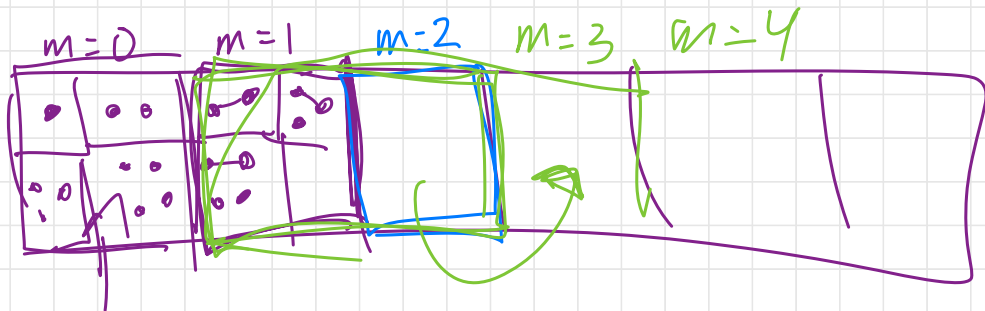
$$= 2m - 2 + 2$$

$$= 2m$$

by IH
since G'
has fewer edges
than G

So, the sum of degrees of vertices of G

$$= 2m.$$



base
case

Recursive Multiplication

$x = 28088990$

$y = 26715635$

$n = 8$

big #

$\text{multiply}(x, y, n):$

if $n = 1$:
return ^{single digit + multiplication} $\text{single digit} + \text{multiply}(x, y)$

if $n > 1$:

$a = \text{first } n/2 \text{ digits of } x$

$b = \text{last } n/2 \text{ digits of } x$

$c =$

$d =$

28080000

$x \cdot y =$

$$(2808 \cdot 10^4 + 8990) (2671 \cdot 10^4 + 5635)$$

$$= (2808 \cdot 2671 \cdot 10^8 + 2808 \cdot 5635 \cdot 10^4$$

$$+ 8990 \cdot 2671 \cdot 10^4 + 8990 \cdot 5635)$$

$$= \text{multiply}(2808, 2671, 4) \cdot 10^8 +$$

$$\text{multiply}(2808 \cdot 5635, 4) \cdot 10^4$$

